6. Dacă $x_i, y_j, i=\overline{1,n}, j=\overline{1,k}$ sunt puncte distincte să se arate că

$$[x_1,x_2,...,x_n,y_1,y_2,...,y_k;f]=[x_1,x_2,...,x_n;\frac{f(x)}{(x-y_1)(x-y_2)...(x-y_k)}]+$$

$$+[y_1,y_2,...,y_k;\frac{f(x)}{(x-x_1)(x-x_2)...(x-x_n)}]$$

Ideea:

$$\begin{aligned}
[t_1,t_2,...,t_m;f] &= \sum_{i=1}^{m}\frac{f(t_i)}{l'(t_i)}\\
l(t) &= \prod_{k=1}^{m}(t-t_k)
\end{aligned}$$

$$\begin{aligned}
[x_1,...,x_n,y_1,...,y_k;f] &= \sum_{i=1}^{n}\frac{f(x_i)}{l'(x_i)}+\sum_{j=1}^{k}\frac{f(y_j)}{l'(y_j)}\\
l(t) &= \prod_{i=1}^{n}(t-x_i)\prod_{j=1}^{k}(t-y_j)\\
l1(t) &= \prod_{i=1}^{n}(t-x_i), l2(t)=\prod_{j=1}^{k}(t-y_j)\\
l'(x_i) &= l1'(x_i)\,l2(x_i)+l1(x_i)\,l2'(x_i)=\\
&= l1'(x_i)\,l2(x_i)\\
l'(y_j) &= l1(y_j)\,l2'(y_j)
\end{aligned}$$

$$\begin{aligned}
[x_1,...,x_n,y_1,...,y_k;f] &= \sum_{i=1}^{n}\frac{f(x_i)}{l1'(x_i)\,l2(x_i)}+\sum_{j=1}^{k}\frac{f(y_j)}{l1(y_j)\,l2'(y_j)}=\\
&= \sum_{i=1}^{n}\frac{\frac{f(x_i)}{l2(x_i)}}{l1'(x_i)}+\sum_{j=1}^{k}\frac{\frac{f(y_j)}{l1(y_j)}}{l2'(y_j)}=\\
&= \left[x_1,...,x_n;\frac{f}{l2}\right]+\left[y_1,...,y_k;\frac{f}{l1}\right]
\end{aligned}$$

$[x_1,x_2,...,x_n,y_1,y_2,...,y_k;(x-y_1)(x-y_2)...(x-y_k)f(x)]=[x_1,x_2,...,x_n;f]$

unde $x_i\neq y_j, i=\overline{1,n}, j=\overline{1,k}$ şi $x_i\neq x_j$ $y_i\neq y_j$ pentru $i\neq j$.

Se aplică 6. pentru $f(x) := l2(x)\, f(x)$

$$[0, 1, 2, ..., 100; \frac{1}{(x+1)(x+2)}]$$

Aleg în 6. $f(x) = 1$; $x_1 = 0, ..., x_{101} = 100; y_1 = -1, y_2 = -2$.

$$\begin{aligned}
0 &= [0, 1, ..., 100, -1, -2; 1] = \\
&= \left[0, 1, ..., 100; \frac{1}{(y+1)(y+2)}\right] + \left[-1, -2; \frac{1}{x(x-1)...(x-100)}\right] \\
&= \left[0, 1, ..., 100; \frac{1}{(y+1)(y+2)}\right] + \frac{\frac{1}{\prod_{i=0}^{100}(-1-i)}}{l2'(-1)} + \frac{\frac{1}{\prod_{i=0}^{100}(-2-i)}}{l2'(-2)}
\end{aligned}$$

$$l2(x) = (x+1)(x+2)$$

$$\frac{\frac{1}{\prod_{i=0}^{100}(-1-i)}}{l2'(-1)} + \frac{\frac{1}{\prod_{i=0}^{100}(-2-i)}}{l2'(-2)} = \frac{(-1)^{101}}{101!}/(+1) + \frac{(-1)^{101}}{102!}/(-1)$$

Fie $f \in C[a, b]$. Să se arate că pentru orice $\lambda \in [0, 1]$ avem

$$(1-\lambda)f(a) + \lambda f(b) - f((1-\lambda)a + \lambda b) = k[\theta_1, \theta_2, \theta_3; f]$$

unde $k, \theta_1, \theta_2, \theta_3$ sunt numere independente de funcţia f care se vor determina.

$$\begin{aligned}
k[\theta_1, \theta_2, \theta_3; f] &= \frac{f''(\xi)}{2!} = k\left(\frac{f(\theta_1)}{l'(\theta_1)} + \frac{f(\theta_2)}{l'(\theta_2)} + \frac{f(\theta_3)}{l'(\theta_3)}\right) \\
l(x) &= (x-\theta_1)(x-\theta_2)(x-\theta_3)
\end{aligned}$$

prin comparare:

$$\begin{aligned}
\theta_1 &= a, \theta_2 = b, \theta_3 = ((1-\lambda)a + \lambda b) \\
l(x) &= (x-a)(x-b)(x-((1-\lambda)a + \lambda b)) \\
l'(a) &= (a-b)^2\lambda, l'(b) = (a-b)^2(1-\lambda) \\
l'(((1-\lambda)a + \lambda b)) &= -(a-b)^2(1-\lambda)\lambda
\end{aligned}$$

$$[0, 1, 2, ..., n; \frac{x^{n+1}}{x+1}].$$

Figure 1:

$$\begin{aligned}
k\left[\theta_1, \theta_2, \theta_3; f\right] &= k\left(\frac{f(a)}{(a-b)^2\lambda} + \frac{f(b)}{(a-b)^2(1-\lambda)} + \frac{f((1-\lambda)a + \lambda b)}{-(a-b)^2(1-\lambda)\lambda}\right) = \\
&= (1-\lambda) f(a) + \lambda f(b) - f((1-\lambda)a + \lambda b) \Rightarrow
\end{aligned}$$

$$\begin{aligned}
\frac{k}{(a-b)^2\lambda} &= (1-\lambda) \\
\frac{k}{(a-b)^2(1-\lambda)} &= \lambda \\
\frac{k}{-(a-b)^2(1-\lambda)\lambda} &= -1
\end{aligned}$$

$$k = (a-b)^2(1-\lambda)\lambda$$

cf. 6.

$$\begin{aligned}
\left[0, 1, ..., n, -1; x^{n+1}\right] &= \left[0, 1, ..., n; \frac{x^{n+1}}{x+1}\right] + \left[-1; \frac{x^{n+1}}{x(x-1)...(x-n)}\right] \\
1 &= \left[0, 1, ..., n; \frac{x^{n+1}}{x+1}\right] + \frac{(-1)^{n+1}}{(-1)(-2)...(-1-n)} \\
1 &= \left[0, 1, ..., n; \frac{x^{n+1}}{x+1}\right] + \frac{1}{(n+1)!}
\end{aligned}$$

$$[0, 1, 2, ..., n; x^{n+2}].$$

$$\begin{aligned} l\left(x\right) &= \prod_{i=0}^{n}\left(x-i\right) \\ x^{n+2} &= \left(x+\alpha\right)l(x)+r\left(x\right) \\ r\left(x\right) &= \beta x^{n}+r1\left(x\right) \end{aligned}$$

$$\begin{aligned} \left[0,1,...,n;x^{n+2}\right] &= \left[0,1,...,n;\left(x+\alpha\right)l(x)\right]+ \\ +\beta\left[0,1,...,n;x^{n}\right]+\left[0,1,...,n;r1\left(x\right)\right] &= \\ &= \beta\cdot 1 \end{aligned}$$

$$x^{n+2}=\left(x+\alpha\right)l(x)+\beta x^{n}+r1\left(x\right)$$

coef. x^{n+1} :

$$0=\alpha+coef\ x^{n}din\ l\left(x\right)$$

$$\alpha=0+1+..+n=\frac{n\left(n+1\right)}{2}$$

coef. x^{n} :

$$\begin{aligned} 0 &= coef\ x^{n-1}din\ l\left(x\right)+\alpha coef\ x^{n}din\ l\left(x\right)+\beta \\ 0 &= \sum_{\substack{i=1\\ j=1,i<j}}^{n} ij-\alpha^{2}+\beta=\frac{1}{2}\left(\left(\sum_{i=1}^{n}i\right)^{2}-\sum_{i=1}^{n}i^{2}\right)-\left(\frac{n\left(n+1\right)}{2}\right)^{2}+\beta \end{aligned}$$

:de unde rezultă β.

Să se demonstreze formula (T. Popoviciu):

$$\begin{aligned} \left[x_0,x_1,...,x_n;fg\right] &= \sum_{k=0}^{n}\left[x_0,...,x_k;f\right]\left[x_k,...,x_n;g\right] \\ \text{Leibniz} &: \left(fg\right)^{(n)}=\sum_{k=0}^{n}\binom{n}{k}f^{(k)}g^{(n-k)} \end{aligned}$$

Ideea:

$$\left[x_0,x_1,...,x_n;fg\right]=\left[x_0,x_1,...,x_n;L(f)g\right] \tag{1}$$

unde $L\left(f\right)$ este pol. lui Lagrange pt. f , nodurile $x_0,x_1,...,x_n$, scris sub forma Newton:

$$L\left(f\right)\left(x\right)=f\left(x_0\right)+\sum_{k=1}^{n}\prod_{i=0}^{k-1}\left(x-x_i\right)\left[x_0,..,x_k;f\right]$$

Înlocuind în 1 obţinem:

$$\begin{aligned} [x_0, x_1, ..., x_n; fg] &= f(x_0)[x_0, x_1, ..., x_n; g] + \\ + \sum_{k=1}^{n} [x_0, .., x_k; f] \left[x_0, x_1, ..., x_n; \prod_{i=0}^{k-1} (x - x_i) g \right] &= \\ &= f(x_0)[x_0, x_1, ..., x_n; g] + \\ + \sum_{k=1}^{n} [x_0, .., x_k; f][x_k, ..., x_n; g] &= \sum_{k=0}^{n} [x_0, .., x_k; f][x_k, ..., x_n; g] \end{aligned}$$

S-a folosit:

$[x_1, x_2, ..., x_n, y_1, y_2, ..., y_k; (x-y_1)(x-y_2)...(x-y_k)f(x)] = [x_1, x_2, ..., x_n; f]$

unde $x_i \neq y_j, i = \overline{1,n}, j = \overline{1,k}$ şi $x_i \neq x_j$ $y_i \neq y_j$ pentru $i \neq j$.

$$n := k; k := k; x_1 := x_k; ..., x_n := x_n; y_1 := x_0, ..., y_k := x_{k-1}$$

$$\left[x_0, x_1, ..., x_n; \prod_{i=0}^{k-1} (x - x_i) g \right] = [x_k, ..., x_n; g]$$